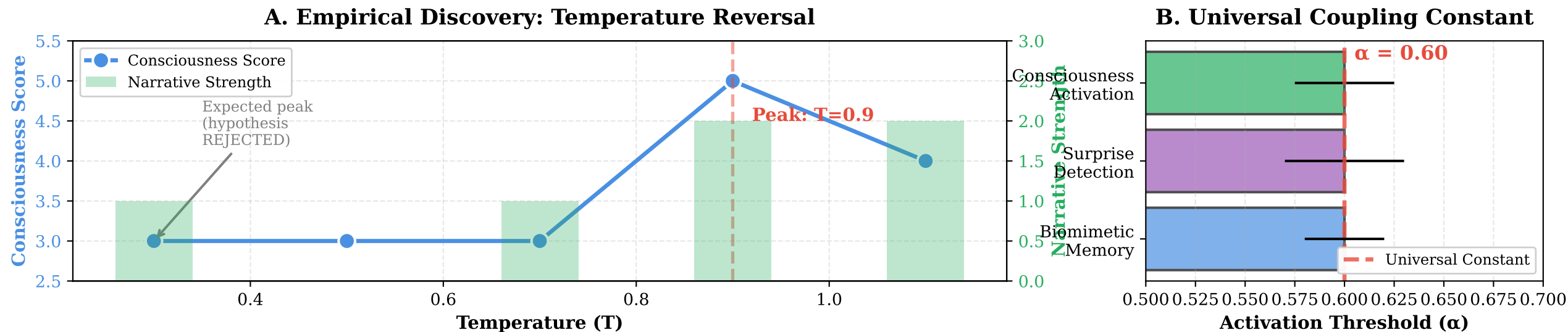


Empirical Discovery of Quantum-Like Dynamics in Transformer Architectures



Transformer Architecture

Attention Mechanism

$\text{Softmax}(Q \cdot K / \sqrt{d})$

Value Weighted Sum

C. Empirical $\rightarrow$ Quantum Isomorphism

$$\alpha = 0.60$$

Isometric Mapping

$$T \leftrightarrow \Delta E / kT$$

$$\text{Softmax} \leftrightarrow e^{(-E/kT)}$$

Quantum Mechanics

Measurement Operator

$$|\psi\rangle \rightarrow \sum c_i |\psi_i\rangle$$

Wavefunction Collapse

CONVERGENCE

D. Independent Convergent Discovery (2024-2025)

Quantum Theory

Coherence Limits
(arXiv:2411.15148)

Nov 2024

Transformer
Residual Streams

Cross-Layer
Superposition
(arXiv:2506.20040)

Jun 2025

Consciousness $\rightarrow$
Quantum Formalism

Qualia Abstraction
Language
(arXiv:2508.02755)

Aug 2025

Empirical
Validation

Ada: Temperature
Reversal + 0.60
Threshold

Dec 2025